

CET 479_ Spring 2023

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In class Activity – March 20, 2023

LAB_07 : Use Python in Linux environment

Use your Linux environment and vi editor for the following:

1. Write an expression to calculate the value of y , where the initial value of y is 0.9 .
Expression is $y = y*y*(1-y)$. Print the value of y .

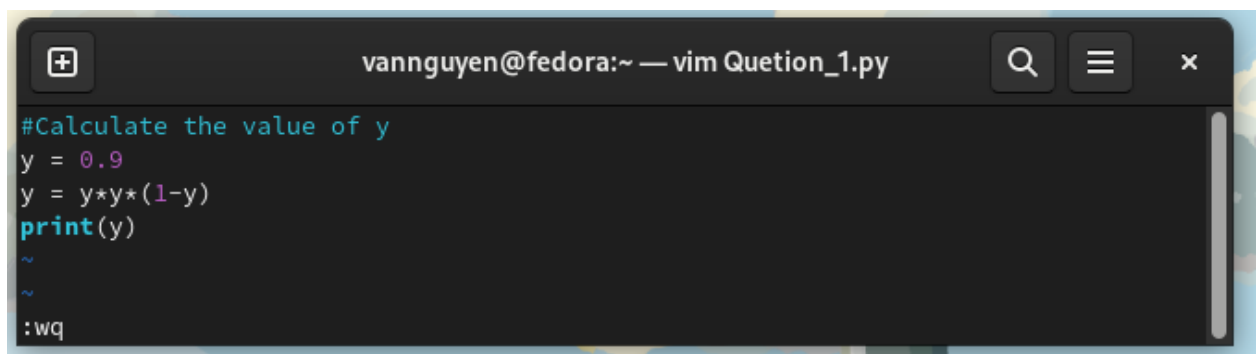
\$ **Vim** **Question_1.py** : Create a file vim editor

(**i** insert)

```
# Calculate the value of y
y = 0.9
y = y*y*(1- y)
print(y)
```

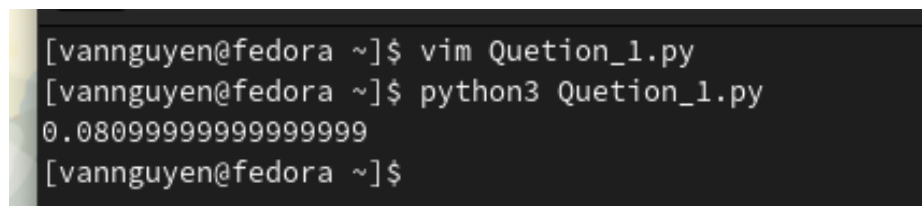
(**esc** :**wq** save and quit)

\$ **python3** **Question_1.py** : run file



```
vannguyen@fedora:~ — vim Quetion_1.py
#Calculate the value of y
y = 0.9
y = y*y*(1-y)
print(y)
~
~
:wq
```

output

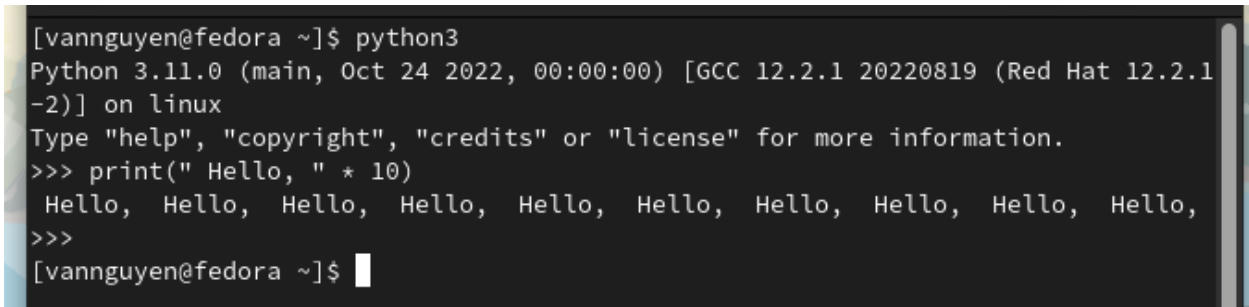


```
[vannguyen@fedora ~]$ vim Question_1.py
[vannguyen@fedora ~]$ python3 Question_1.py
0.08099999999999999
[vannguyen@fedora ~]$
```



2. Write an expression in python to print “Hello” 10 times.

```
$ python3 : call python
>>> print(" Hello, " * 10)
```



```
[vannguyen@fedora ~]$ python3
Python 3.11.0 (main, Oct 24 2022, 00:00:00) [GCC 12.2.1 20220819 (Red Hat 12.2.1-2)] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> print(" Hello, " * 10)
Hello, Hello, Hello, Hello, Hello, Hello, Hello, Hello, Hello, Hello,
>>>
[vannguyen@fedora ~]$
```



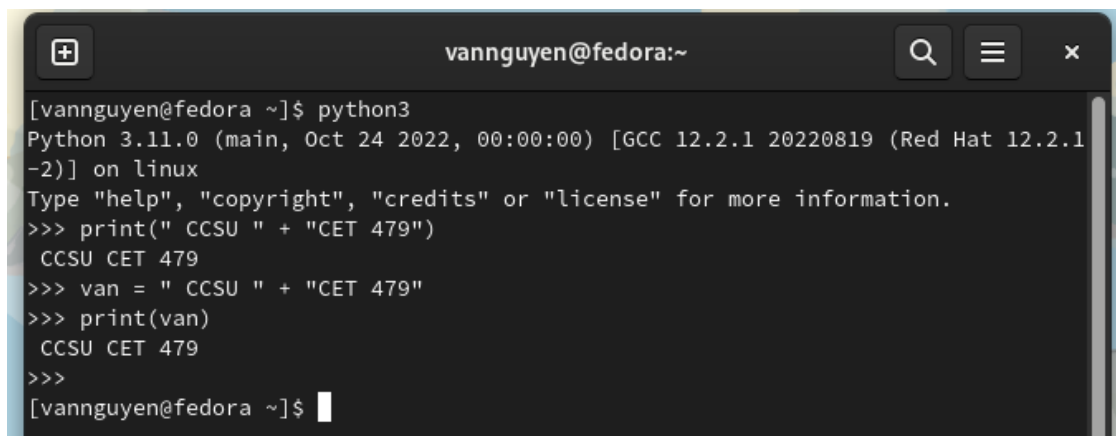
3. Show an example when the + operator is used to concatenate two strings.

```
>>> print("CCSU " + "CET 479 ")
```

Or

```
>>> van = "CCSU " + "CET 479"
>>> print(van)
```

Ctrl+D : quit



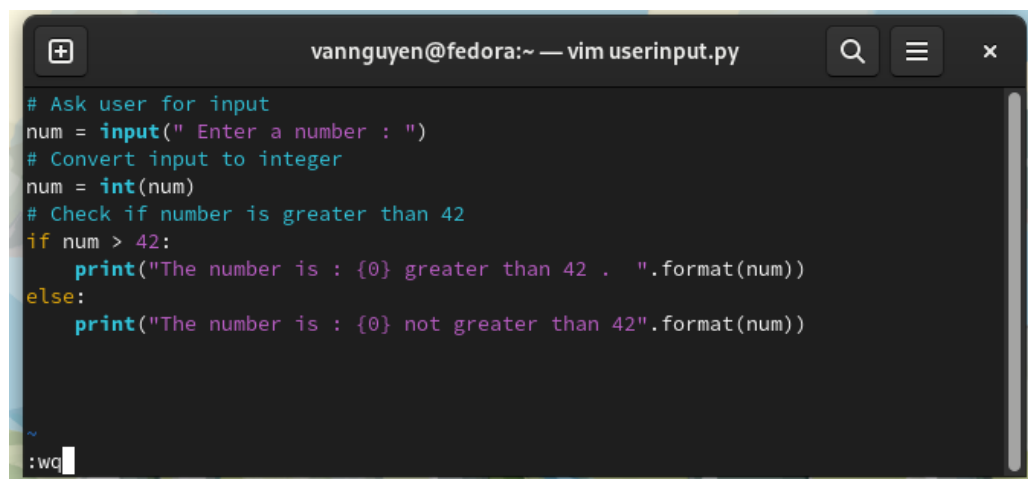
```
vannguyen@fedora:~
[vannguyen@fedora ~]$ python3
Python 3.11.0 (main, Oct 24 2022, 00:00:00) [GCC 12.2.1 20220819 (Red Hat 12.2.1-2)] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> print(" CCSU " + "CET 479")
CCSU CET 479
>>> van = " CCSU " + "CET 479"
>>> print(van)
CCSU CET 479
>>>
[vannguyen@fedora ~]$
```



4. Write a code to take an input from the user and check if that number is greater than 42.

```
$ vim userInput.py          : Create a file Vi editor ".py"

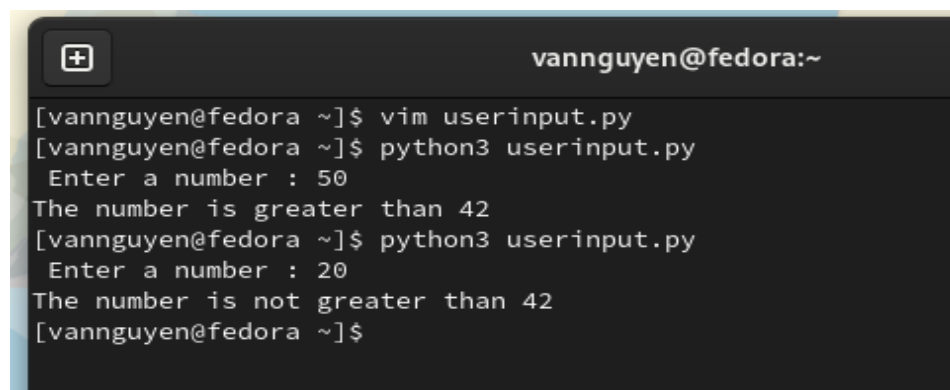
# Ask user for input
num = input("Enter a number : " )
# Convert input to integer
num = int(num)
# Check if number is greater than 42
If num > 42:
    Print("The number is greater 42")
Else:
    Print(" The number is not greater than 42 ")
```

A screenshot of a terminal window titled 'vannguyen@fedora:~ — vim userInput.py'. The window shows the Python code from the previous block. At the bottom, the command ':wq' is entered, indicating the file has been saved and the editor has been exited.

```
vannguyen@fedora:~ — vim userInput.py
# Ask user for input
num = input(" Enter a number : ")
# Convert input to integer
num = int(num)
# Check if number is greater than 42
if num > 42:
    print("The number is : {0} greater than 42 . ".format(num))
else:
    print("The number is : {0} not greater than 42".format(num))

~
:wq
```

Output

A screenshot of a terminal window titled 'vannguyen@fedora:~'. It shows the execution of the 'userInput.py' script twice. In the first run, the user enters '50' and the output is 'The number is greater than 42'. In the second run, the user enters '20' and the output is 'The number is not greater than 42'.

```
vannguyen@fedora:~
[vannguyen@fedora ~]$ vim userInput.py
[vannguyen@fedora ~]$ python3 userInput.py
Enter a number : 50
The number is greater than 42
[vannguyen@fedora ~]$ python3 userInput.py
Enter a number : 20
The number is not greater than 42
[vannguyen@fedora ~]$
```



5. Write a program to display if a number is even or odd. You can assume a number or take it as an input from the user.

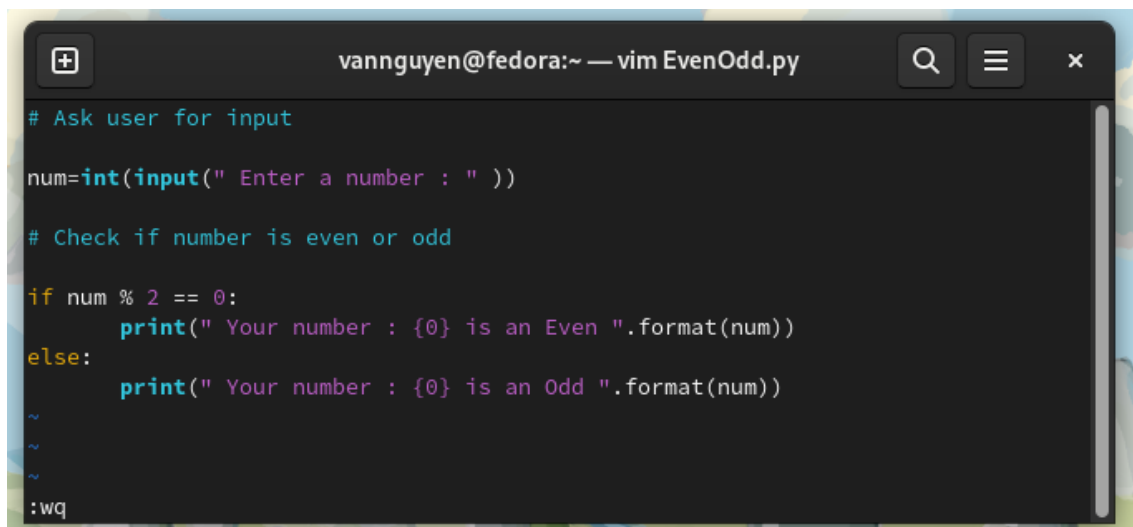
```
$ vim EvenOdd.py          : Create a file Vi editor ".py"
                                                                    (i:inset)

# Ask user for input

num = int(input("Enter your number: "))

# Check if number is even or odd

if num % 2 == 0:
    print("Your number : {0} is an even".format(num))
else:
    print("Your number : {0} is an odd".format(num))
```

A screenshot of a terminal window with a dark background. The title bar shows 'vannguyen@fedora:~ — vim EvenOdd.py'. The terminal displays the Python code from the previous block, with syntax highlighting. At the bottom, the command ':wq' is entered, indicating the end of the file editing session.

Output :

```
[vannguyen@fedora ~]$ vim EvenOdd.py
[vannguyen@fedora ~]$ ls -l EvenOdd.py
-rw-r--r--. 1 vannguyen vannguyen 236 Mar 25 07:01 EvenOdd.py
[vannguyen@fedora ~]$ python3 EvenOdd.py
Enter a number : 15
Your number : 15 is a Odd
[vannguyen@fedora ~]$ python3 EvenOdd.py
Enter a number : 20
Your number : 20 is a Even
[vannguyen@fedora ~]$
```



6. Write a pay computation program that gives the employee 1.5 times the hourly rate for hours worked above 40 hours (and regular 1.0 rate for less than 40 hours). The hours and rate should be taken as input from the user.

Enter Hours : 45

Enter Rate : 10

Pay: 475.0

$$475 = 40 * 10 + 5 * 15$$

Solution:

```
$ vim pay_Computation.py
```

(i : insert)

```
# Ask user for input
```

```
hours = int(input(" Enter Hours : "))
```

```
rate = int(input(" Enter Rate : " ))
```

```
# Calculate for pay Computation
```

```
If hours > 40:
```

```
    pay = (40*rate)+(hours-40)*15
```

```
Else:
```

```
    pay = hours*rate
```

```
# Print result
```

```
print(f" Pay : ${pay:.2f}")
```

(esc exit, :wq save and quit)





```
vannguyen@fedora:~ — vim Pay_Computation.py
# Ask user for input

hours=int(input(" Enter Hours  : " ))
rate=int(input(" Enter Rate   : " ))

# Calculate for Pay Computation

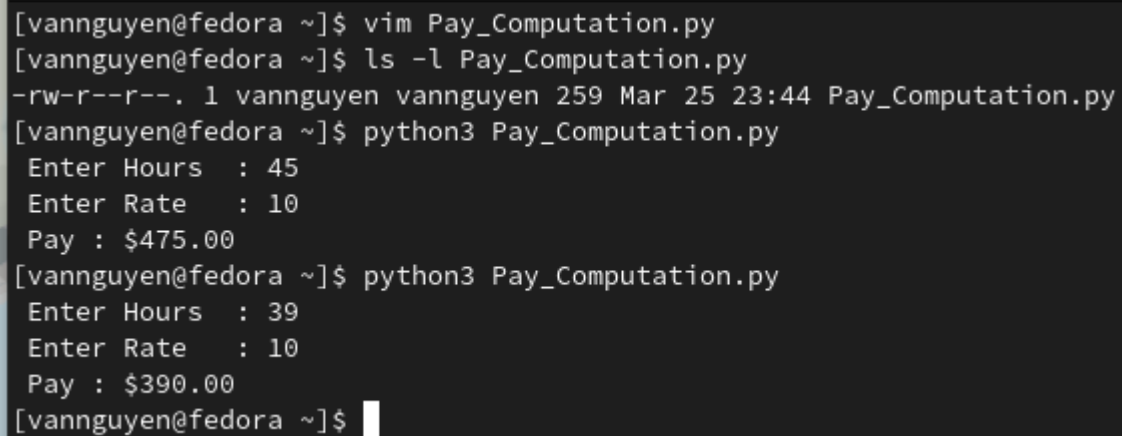
if hours > 40:
    pay = (40*rate)+(hours-40)*15
else:
    pay = hours*rate

# Print result

print(f" Pay : ${pay:.2f}")

:wq
```

Output :



```
[vannguyen@fedora ~]$ vim Pay_Computation.py
[vannguyen@fedora ~]$ ls -l Pay_Computation.py
-rw-r--r--. 1 vannguyen vannguyen 259 Mar 25 23:44 Pay_Computation.py
[vannguyen@fedora ~]$ python3 Pay_Computation.py
Enter Hours  : 45
Enter Rate   : 10
Pay : $475.00
[vannguyen@fedora ~]$ python3 Pay_Computation.py
Enter Hours  : 39
Enter Rate   : 10
Pay : $390.00
[vannguyen@fedora ~]$
```

7. (Rewrite your pay program using try and except so that your program handles non-numeric input gracefully.

Enter Hours: 20
Enter Rate: nine
Error, please enter numeric input

Enter Hours: forty
Error, please enter numeric input

Solution:

```
$ vim pay_Computation.py

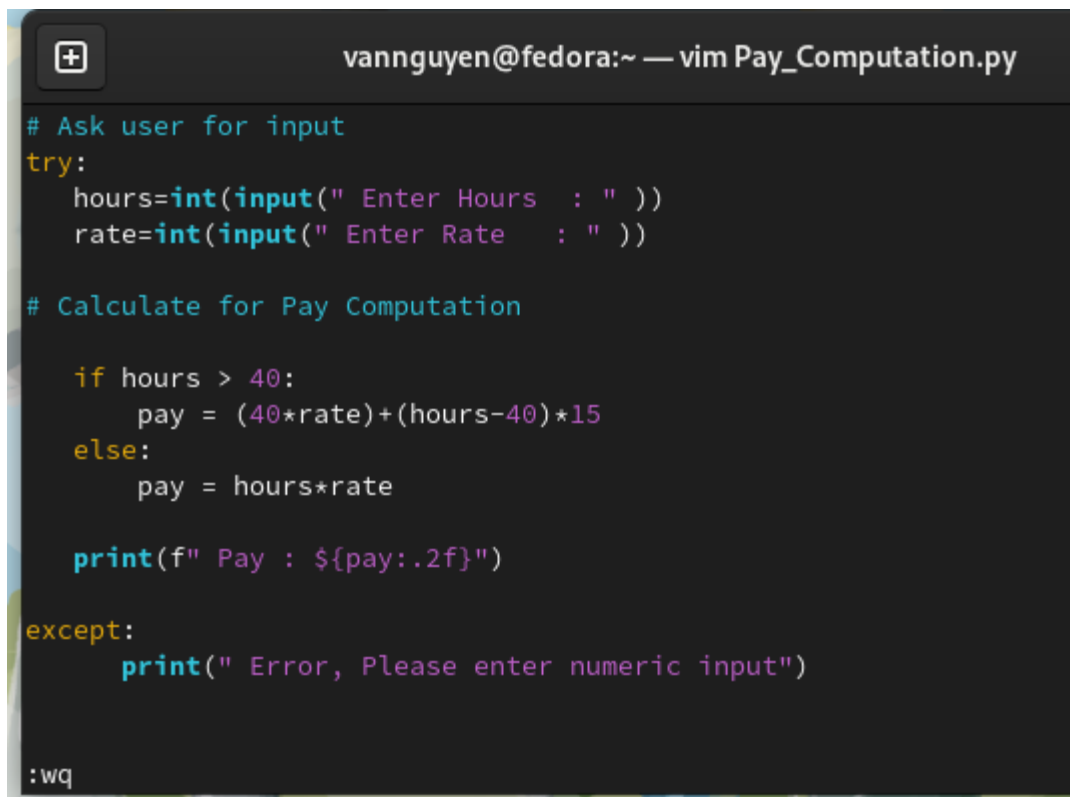
try:

    hours = int(input(" Enter Hours : "))
    rate = int(input(" Enter Rate : " ))
    If hours > 40:
        pay = (40*rate)+(hours-40)*15
    Else:
        print(f" Pay : ${pay:.2f}")

except:

    print(" Error, Please enter numeric input ")
```





```
vannguyen@fedora:~ — vim Pay_Computation.py
# Ask user for input
try:
    hours=int(input(" Enter Hours  : " ))
    rate=int(input(" Enter Rate   : " ))

# Calculate for Pay Computation

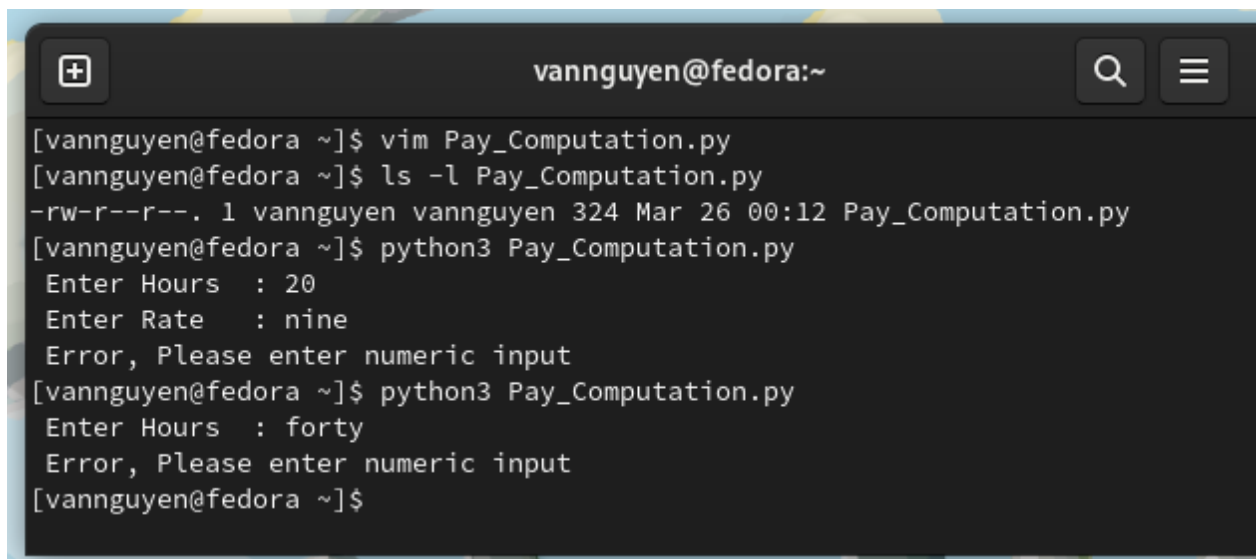
    if hours > 40:
        pay = (40*rate)+(hours-40)*15
    else:
        pay = hours*rate

    print(f" Pay : ${pay:.2f}")

except:
    print(" Error, Please enter numeric input")

:wq
```

Output :



```
vannguyen@fedora:~
[vannguyen@fedora ~]$ vim Pay_Computation.py
[vannguyen@fedora ~]$ ls -l Pay_Computation.py
-rw-r--r--. 1 vannguyen vannguyen 324 Mar 26 00:12 Pay_Computation.py
[vannguyen@fedora ~]$ python3 Pay_Computation.py
Enter Hours  : 20
Enter Rate   : nine
Error, Please enter numeric input
[vannguyen@fedora ~]$ python3 Pay_Computation.py
Enter Hours  : forty
Error, Please enter numeric input
[vannguyen@fedora ~]$
```

